

The physics of the fall

Diagnosing the mechanical and systemic failure of liquid loading in gas wells.

Reservoir depletion turns gas extraction into a fight against gravity

The Healthy State

High-pressure gas wells possess enough kinetic energy to produce gas that easily carries liquid water or condensate upward in the form of a fine mist.

The Critical Threshold

The carrying capacity of the gas steadily decreases until it crosses a critical velocity threshold, allowing gravity to overtake upward momentum.

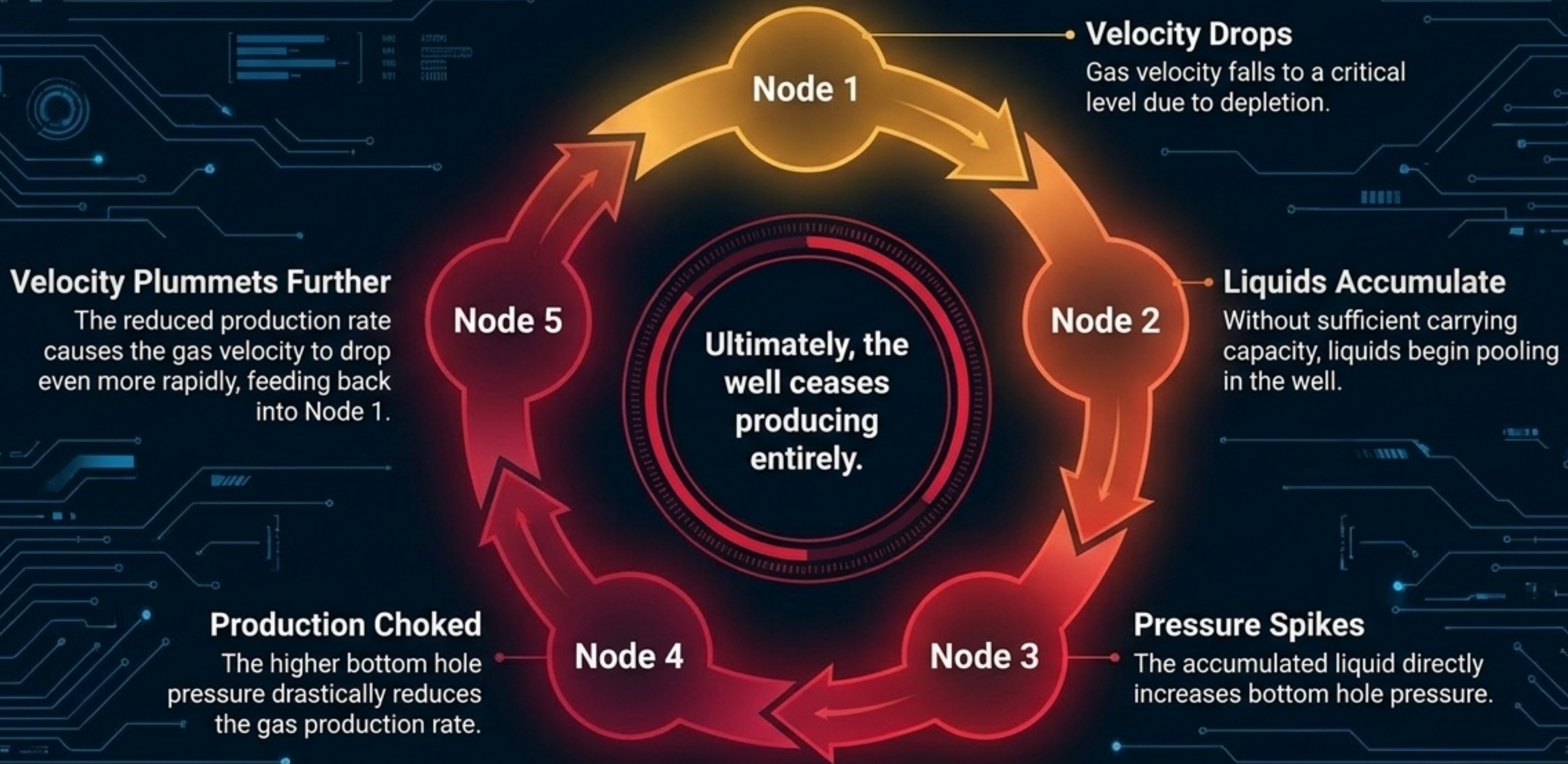
GRAVITY

The Catalyst

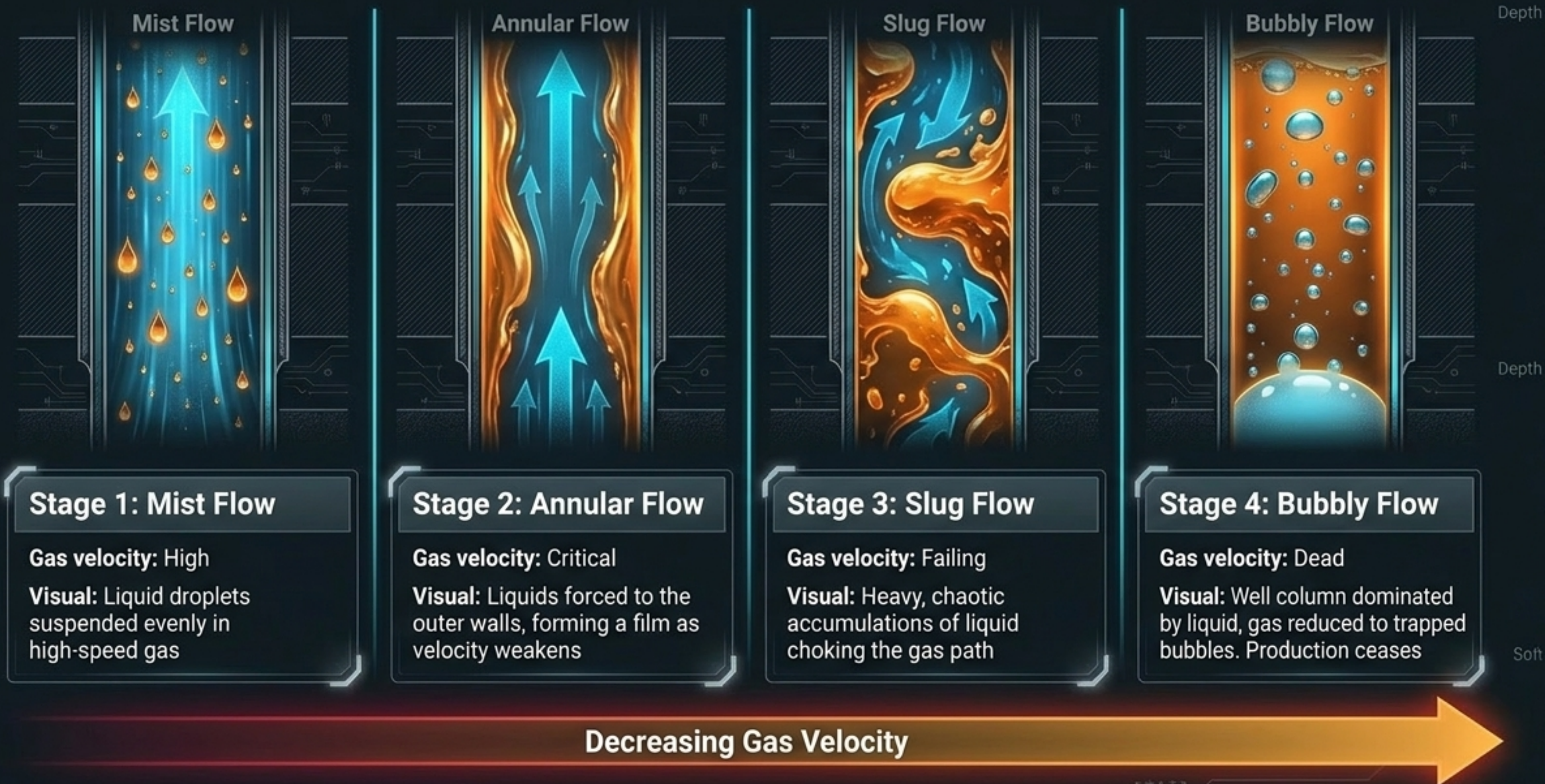
As reservoir pressure depletes over time, the gas flow velocity inevitably drops.

GAS CARRYING
CAPACITY

Liquid loading triggers a vicious, self-reinforcing death spiral



The anatomy of decline maps to four distinct flow regimes



The symptoms of accumulation are dangerously delayed at the surface

Subsurface Realities



Recognizing liquid loading is not an easy task. Thorough diagnostic analysis is required because the problem is not always visually obvious.

Diagnostic Forensics Checklist (Delayed Surface Observations)

- ☒ Onset of liquid slugs arriving at the surface.
- ☒ Increasing difference between tubing and casing pressures over time.
- ☒ Sharp changes in the gradient on a flowing pressure survey.
- ☒ Sharp, anomalous drops in the production decline curve.

Accurate prediction requires modeling multi-phase bottom hole pressures

Taking timely measures requires analytical prediction before surface symptoms appear. However, historical methods often show discrepancies due to the immense difficulty of predicting bottom hole pressure in multi-phase flow environments.

The Pioneer Models (Turner, Hubbard, and Dukler, 1969)

These investigators pioneered the analysis and prediction of the minimum gas flow rate capable of removing liquids.

Model A

The Film Movement Model



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Model B

The Entrained Drop Movement Model

